

DEYUAN (MIKE) HE

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Formal Methods × LLM Agents — verified code generation, specification learning, and provably safe LLM agents.

EDUCATION

PRINCETON UNIVERSITY, Princeton, NJ 2022 – 2027 (expected)

Ph.D. in Computer Science · Advisor: Prof. Aarti Gupta

UNIVERSITY OF WASHINGTON, Seattle, WA 2018 – 2022

B.S. in Computer Science · Advisors: Prof. Zachary Tatlock & Dr. Steven Lyubomirsky

SELECTED PUBLICATIONS (full list on [Google Scholar](#))

- **SPECY: Learning Specifications for Distributed Systems from Event Traces**
Mike He, A. Desai, A. Jagarapu, D. Terry, S. Malik, A. Gupta. *OOPSLA*, 2026.
- **SOTERIA: Formally Verified Planning with Runtime Enforcement for Safe LLM Agents**
Mike He, A. Desai, S. Malik, A. Gupta. *Under submission*, 2026.
- **ALGOVERI: An Aligned Benchmark for Verified Code Generation on Classical Algorithms**
H. Zhao*, Z. Yang*, J. Li*, Mike He*, Z. Li*, C. Jin, V. Veeravalli, A. Gupta, S. Arora. *ICML*, 2026. (Spotlight)
- **Goedel-Code-Prover: Hierarchical Proof Search for Open State-of-the-Art Code Verification**
Z. Li*, Z. Yang*, Mike He, H. Zhao, A. Zhao, S. Tang, K. Yang, A. Gupta, Z. Su, C. Jin. *COLM*, 2026. *: equal contribution.

RESEARCH EXPERIENCE

Princeton University · Graduate Research Assistant 2022 – Present

- **SOTERIA** — verified planning & runtime enforcement for LLM agents (first author, in submission).
A Python-like plan DSL, a verification-condition generator for temporal and data-flow properties, and a runtime monitor binding execution to the plan. *Result*: zero policy violations over **207 tasks** (τ^2 -bench, Toolathon), 4 LLM-backed agents.
- **ALGOVERI** — cross-language vericoding benchmark (co-first author; ICML 2026 Spotlight).
Aligned formal specifications for 77 classical algorithms in Dafny, Verus, and Lean. *Result*: best-model success collapses from **40.3%** (Dafny) to **7.8%** (Lean), isolating verification-language design as the dominant factor.
- **Goedel-Code-Prover** — LLM-guided proof search for code verification (COLM 2026).
Curated the training corpus; co-designed losses exploiting proof context and Lean verifier feedback. *Result*: the 8B model proves **62.0%** of 427 tasks — $2.6\times$ the best baseline, beating frontier models (18.5%) and proves $84\times$ larger.

Amazon Web Services, Agentic Automated Reasoning · Applied Scientist Intern Summers 2024 – 2026

- **SPECY** — specification learning for distributed systems (first author; OOPSLA 2026).
Designed an event-based specification grammar and a learner combining enumerative search with dynamic trace learning. *Result*: learned **all 43 specifications** across 14 protocols (28 fully automatically), recovered hand-written invariants, and exposed a test-coverage gap in a production AWS service.
- **PVERIFIER** — deductive verification of P protocols; shipped in the public release.
Designed **ProofScript**, a DSL for proof decomposition. *Result*: proved safety properties (consensus, atomicity) for ring leader election, Two-Phase Commit, and an internal AWS service.
- **Strata** — unified verification framework in Lean: built the concurrency primitives for modeling asynchronous programs; enabled compositional verification for asynchronous Python.

Earlier Industry Research 2022

- **Taichi Graphics, Compiler R&D Intern** — added a first-class tensor type to the Taichi IR and adapted its dataflow passes.
- **Intel Strategic CAD Lab, Formal Verification Intern** — PYROPE: correct-by-construction hardware modeling in Dafny; verified multi-Montgomery reduction.

TECHNICAL SKILLS

Formal methods: Lean 4, Dafny, P, Z3/SMT, specification inference, equality saturation

LLMs & agents: coding agents, vericoding, autoformalization, LLM-guided proof search, runtime enforcement

Programming: Python, Java, C#, Rust · **Systems**: distributed systems, compilers, program analysis

HONORS, TEACHING & SERVICE

Honors: Spotlights — ICML'26 ($\times 2$), ICLR'21 · CRA Outstanding Undergrad. Researcher, Honorable Mention (2022)

Teaching: TA — COS 516: Automated Reasoning about Software (Princeton, $\times 2$); CSE 505 (UW)

Service: Reviewer — NeurIPS, IEEE TMC · Artifact Evaluation — POPL'25, PLDI'24, POPL'24, MLSys'23, MICRO'21